

## Multiple Choice (10 marks)

1. A group of students observed the formation of bubbles when water boiled. What happened to matter because of the physical change to the water?
  - (a) New matter was created in the bubbles.
  - (b) Old matter was destroyed as the water turned to gas.
  - (c) Matter was conserved when it changed form.
  - (d) The composition of the matter changed.
2. Why does sexual reproduction result in more genetic diversity than asexual reproduction?
  - (a) Traits from two parents are combined.
  - (b) More organisms reproduce this way.
  - (c) Offspring grow in different environments.
  - (d) Offspring come from identical parents.
3. Paramecium and volvox can both reproduce sexually. How is sexual reproduction in volvox different than sexual reproduction in paramecium?
  - (a) Volvox colonies produce gametes.
  - (b) Volvox cells undergo conjugation.
  - (c) Volvox colonies produce single-celled offspring.
  - (d) Volvox cells reproduce by vegetative budding.
4. The human population is increasing every day. How does this increase in population affect the natural resources that maintain humans?
  - (a) increases the supply of resources
  - (b) decreases the amount of resources
  - (c) results in less demand for resources
  - (d) improves the quality of the resources
5. All of these are examples of the ways Earth and the Moon interact except
  - (a) the phases of the Moon.
  - (b) the tides on Earth.
  - (c) seasons on Earth.
  - (d) lunar eclipses.

## True / False (10 marks)

*Answer True or False*

6. True or False: Talons in hawks are an acquired characteristic developed through their hunting experiences.
7. True or False: The first action a student should take after a spill is to notify the teacher for proper handling of the situation.
8. True or False: The new organism could belong to the kingdoms of Protists and Plants due to its multicellular and autotrophic characteristics.
9. True or False: Balloons floating upward from the same height show how gravity affects objects in motion.
10. True or False: If the data do not support the hypothesis, it is important to check for errors and repeat the experiment.

## Long Form Response (20 marks)

*Answer each question with a clear and complete explanation.*

11. Discuss the process that occurs when a plastic rod is rubbed with fur, explaining how this interaction results in the rod acquiring a negative charge.
12. Discuss how the communication process within cells, where molecules relay messages from the membrane to the nucleus, parallels the functioning of a specific body system.

*End of Paper*